

Best Practices for Writing Scientific Paper Introductions

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Background and Purpose of Scientific Introductions

Scientific introductions serve as expanded versions of abstracts that provide context, motivation, and a clear argument for why the research is important. They should be self-contained, brief, and focus on building a compelling case for the study rather than providing comprehensive historical reviews. (4 sources)

The introduction of a scientific paper serves as a bridge between the abstract and the main content, essentially presenting "the same content (summary of the article), in the same order, as the abstract, with more space provided" and typically being "about ten times longer as a rough approximation" ([Stojmenovic et al., 2012](#)). The primary purpose is to create a self-contained section that allows "important readers and evaluators" to understand the contribution and validity of the work without needing to read the entire manuscript ([Stojmenovic et al., 2012](#)).

A well-crafted introduction should "outline the general context of the study, and provides the reader with a swift understanding of the paper content" while its "main function is to motivate the paper, e.g., by explaining why the problem to be solved is important" ([Verdecchia et al., 2025](#)). The key is building "a clear chain of arguments from the context via the problem to your objective, all supported by appropriate references" ([Verdecchia et al., 2025](#)). Rather than simply stating that nobody has done the work before, authors should focus on "why is it important to do it, and what do we have to gain from it" ([Verdecchia et al., 2025](#)).

Importantly, introductions should not function as comprehensive literature reviews. The "most important and most neglected characteristic of the introduction is that it must not be a review" ([Neto et al., 2023](#)). Instead of chronological narratives starting from historical discoveries, effective introductions should "briefly contextualize -in a few paragraphs, let us say 2 to 4 -the science and the problem faced, showing how important this specific study is to the field" ([Neto et al., 2023](#)). The best introductions create a "magic" effect that captivates readers and motivates engagement with the manuscript ([Neto et al., 2023](#)).

The introduction should also demonstrate thorough understanding of existing work by including "information from systemic review of work already done before and also demonstrate that what more work needs to be done" ([Jawaid et al., 2019](#)). Authors should select "only a few latest studies pertaining to mostly last five years" and ensure that "four to five or at the most one third of the references are covered in the introduction" if the manuscript contains about twenty-five total references ([Jawaid et al., 2019](#)).

Evidence

(Stojmenovic et al., 2012)

How to write research articles in computing and engineering disciplines

"In brief, the introduction of a paper intended to be published in a journal or conference, or Chapter 1 of a master or doctoral thesis, should present the same content (summary of the article), in the same order, as the abstract, with more space provided. Normally it is about ten times longer as a rough approximation"

"In our view, the introduction or Chapter 1 should present sufficient information, and be sufficiently self-contained, so that important readers and evaluators do not need to read the rest of text, being assured in the contribution made, and validity of the text to follow."

(Verdecchia et al., 2025)

Notes On Writing Effective Empirical Software Engineering Papers: An Opinionated Primer

"The introduction serves to outline the general context of the study, and provides the reader with a swift understanding of the paper content. In addition to that, its main function is to motivate the paper, e.g., by explaining why the problem to be solved is important. A reader should ideally believe in this before they reach your stated objective, so build a clear chain of arguments from the context via the problem to your objective, all supported by appropriate references. Furthermore, do not only say that nobody has done it before. There are a great many things that nobody has done before, and it is often better that way. Why is it important to do it, and what do we have to gain from it?"

"A typical Introduction may be composed of the following paragraphs, but especially the later ones are not all mandatory: • Concise introduction to the topic, which explains the need and relevance of the paper. This first part can mention some related work if needed to describe the research context, while leaving a proper discussion of the related work to the dedicated section. • A very brief description of the main contribution of the research, e.g., a new approach, theory, conducted empirical study, or replication. • A brief mention of the used research method(s), allowing the reader to gain a high-level overview of how the study was conducted. • A brief summary of the main results of the work, if possible reporting explicitly the results with precise numbers (e.g., "the approach improves X by Y% "). • A bullet list summarizing all major contributions of the work (e.g., a novel approach, a conducted evaluation experiment, and a reusable dataset)."

(Neto et al., 2023)

The Art of Scientific Writing and Ethical Use of Artificial Intelligence

"The most important and most neglected characteristic of the introduction is that it must not be a review. Some manuscripts have such long introductions with so many distractions that readers -those whom you worked so hard to attract in the title and abstract - quit reading the manuscript while struggling through the introduction"

"beginners often believe they should start introductions with "J. J. Thomson's discovery of the electron..." and go on to craft a chronological narrative until they have fully explained the current knowledge in the field. But the best introductions, rather than providing such historical reviews, should briefly contextualize -in a few paragraphs, let us say 2 to 4 -the science and the problem faced, showing how important this specific study is to the field"

"well-written introductions should do more: they should have a "magic" effect on readers, making them feel captivated by the science and the advancements that the study promises to offer, thus motivating overall engagement with the manuscript"

"Regarding length, introductions should never be more than a few paragraphs (2 or 3 at least and 6 or 7 at most), regardless of the type of manuscript. But the length also depends on many factors and may vary accordingly. If you are writing a full paper in an interdisciplinary field -for instance, when you have monitored a multicomponent reaction with a new mass spectrometric technique - you likely need to write about different areas and concepts; therefore, a longer introduction is justified."

(Jawaid et al., 2019)

How to write introduction and discussion

"It is essential that the introduction is brief, it should be well referenced. It should provide an outline of the current thinking, identify the gaps which exist in knowledge which the author wishes to fill. (1973) Some basic requirements for writing introduction include that it is not only short but also arresting. It should tell the readers what to say, is it worth saying. Is it the right format for conveying this message? Tell the readers why you have undertaken to conduct this study. What is already known on this subject and what new information will be added to the medical literature. (Goldreich, 2000)"

"The best introduction should include information from systemic review of work already done before and also demonstrate that what more work needs to be done. The objective is to answer a question and one should undertake the study only if the question cannot be answered from the available literature. After doing proper literature search, select only a few latest studies pertaining to mostly last five years. Make sure that you have read important studies already undertaken on that particular subject and they should be referenced in this section. If there are about twenty five references in the whole manuscript, make sure that four to five or at the most one third of the references are covered in the introduction. The possible start for an introduction can be an interesting story, vivid description of a scene, using a strong quotation or giving some intriguing facts or making some controversial statements. [4]

Introduciton must also describe the magnitude of the problem which should be followed by brief description of current knowledge and the gap which still exists. [5] At the end of the introduciton, restate the objective of the study."

Core Elements and Structure

Effective scientific introductions follow a structured three-move pattern: establishing the research territory, identifying a knowledge gap, and presenting how the current work fills that gap. They should progress from general context to specific research objectives while answering key questions about the problem, previous work, limitations, and proposed solutions. (6 sources)

The structure of scientific introductions follows well-established rhetorical patterns that guide readers through a logical progression of ideas. Research across multiple disciplines has identified three core rhetorical moves that introductions should follow: "1. establishing a research territory (including, e.g., the step 'reviewing previous research'), 2. identifying a niche (including, e.g., the step 'highlighting a problem or a gap in the existing knowledge'), and 3. occupying the niche (including, e.g., the step 'presenting the contribution of the article itself in addressing that niche')" ([Knight et al., 2020](#)) ([Lam et al., 2012](#)). The language used in each move is particularly important, with phrases like "Prior work has not yet established..." serving to highlight gaps and "In this paper we describe..." introducing the current research ([Knight et al., 2020](#)).

A well-structured introduction should systematically answer several fundamental questions: "what is the problem to be solved? What previous work has been done? What is its main limitation? What do you hope to achieve? How do you set up your investigation?" ([Zelst et al., 2022](#)). The introduction should also "outline the planned theoretical contributions of the paper in relation to previous research" ([Cerpa et al., 2019](#)).

For practical implementation, introductions can follow a structured approach that moves from general to specific context. A typical introduction may include: "• Concise introduction to the topic, which explains the need and relevance of the paper • A very brief description of the main contribution of the research • A brief mention of the used research method(s) • A brief summary of the main results of the work, if possible reporting explicitly the results with precise numbers • A bullet list summarizing all major contributions of the work" ([Verdecchia et al., 2025](#)).

Best practices for structuring introductions include beginning "by providing a concise background account of the problem studied" followed by stating "the objective of the investigation" as "the most important part of the introduction" ([Haswani et al., 2023](#)). Authors should then establish the significance of their work, introduce pertinent literature without providing "a full history of the topic," clearly state hypotheses and methods, and describe major findings and their contributions to the field ([Haswani et al., 2023](#)). The structure should avoid subsections within the introduction and maintain focus on information directly relevant to the research question ([Haswani et al., 2023](#)).

Evidence

(Knight et al., 2020)

[Are You Being Rhetorical? A Description of Rhetorical Move Annotation Tools and Open Corpus of Sample Machine-Annotated Rhetorical Moves](#)

"By analyzing articles across multiple disciplines, Swales (1990) identified three rhetorical moves in introductions, which follow distinct patterns and a series of functional strategies within each move, called "steps":

1. establishing a research territory (including, e.g., the step "reviewing previous research"), 2. identifying a niche (including, e.g., the step "highlighting a problem or a gap in the existing knowledge"), and 3. occupying the niche (including, e.g., the step "presenting the contribution of the article itself in addressing that niche"). Appropriate language use is particularly important to accomplishing these steps and moves. For example, the phrase "Prior work has not yet established..." serves to highlight a gap, and "In this paper we describe..." introduces the present research."

(Lam et al., 2012)

[A comparative study of rhetorical moves in abstracts of published research articles and students' term papers in the field of](#)

[computer and communication systems engineering.](#)

"Swales (1981, 1990) posits that for a writer to write the introduction section of a research article succinctly, he has to write commencing with Move 1 which is establishing a territory. Within Move 1, the writer has to claim centrality to his work, making topic generalization and reviewing previous research"

"Next, he has to establish a niche (Move 2) where the writer has to indicate the gap of his research and finally concluding with Move 3 which is occupying a niche where the writer announces the purpose of his study."

(Zelst et al., 2022)

[101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth](#)

"A good introduction should answer the following questions: what is the problem to be solved? What previous work has been done? What is its main limitation? What do you hope to achieve? How do you set up your investigation? One major mistake is to attempt to do an extensive literature review in the introduction, which often goes off-topic. The introduction serves as the stage to lay out the motivation for the study, and any background reading should focus on the question being addressed."

(Cerpa et al., 2019)

[Editorial: What is Expected of your Paper Manuscript Submission?](#)

"The introduction should outline the planned theoretical contributions of the paper in relation to previous research."

(Verdecchia et al., 2025)

[Notes On Writing Effective Empirical Software Engineering Papers: An Opinionated Primer](#)

"The introduction serves to outline the general context of the study, and provides the reader with a swift understanding of the paper content. In addition to that, its main function is to motivate the paper, e.g., by explaining why the problem to be solved is important. A reader should ideally believe in this before they reach your stated objective, so build a clear chain of arguments from the context via the problem to your objective, all supported by appropriate references. Furthermore, do not only say that nobody has done it before. There are a great many things that nobody has done before, and it is often better that way. Why is it important to do it, and what do we have to gain from it?"

"A typical Introduction may be composed of the following paragraphs, but especially the later ones are not all mandatory: • Concise introduction to the topic, which explains the need and relevance of the paper. This first part can mention some related work if needed to describe the research context, while leaving a proper discussion of the related work to the dedicated section. • A very brief description of the main contribution of the research, e.g., a new approach, theory, conducted empirical study, or replication. • A brief mention of the used research method(s), allowing the reader to gain a high-level overview of how the study was conducted. • A brief summary of the main results of the work, if possible reporting explicitly the results with precise numbers (e.g., "the approach improves X by Y% "). • A bullet list summarizing all major contributions of the work (e.g., a novel approach, a conducted evaluation experiment, and a reusable dataset)."

(Haswani et al., 2023)

[Integrating extensive and intensive reading worksheets on digital platforms](#)

"Keep your Introduction to be very concise, well structured, and inclusive of all the information needed to follow the development of your findings. Do not over-burden the reader by making the introduction too long. Get to the key parts of the paper sooner rather than later. Best Practice: (1) begin the introduction by providing a concise background account of the problem studied; (2) state the objective of the investigation. Your research objective is the most important part of the introduction; (3) establish the significance of your work: why was there a need to conduct the study?; (4) introduce the reader to the pertinent literature. Do not give a full history of the topic. Only quote previous work having direct bearing on the present problem; (5) clearly state your hypothesis, the variables investigated, and concisely summarize the methods used; (6) define any abbreviations or specialized/regional terms; (7) provide a concise discussion of the results and findings of other studies so the reader understands the big picture; (8) describe some of the major findings presented in your manuscript and explain how they contribute to the larger field of research; (9) state the principal conclusions derived from your results, and; (10) identify any questions left unanswered and any new questions generated by your study. Be concise and aware of who will be reading your manuscript and make sure the Introduction is directed to that audience. Move from general to specific: from the problem in the real world to the literature to your research. Last, please avoid to make a sub section in Introduction."

Writing Style and Approach

Scientific introductions should be written for generalist readers with compelling, distinctive voice while maintaining brevity and clarity. The writing should follow reader expectations by placing old information at sentence beginnings and new information at the end, with paragraphs serving as coherent units of composition. (5 sources)

Scientific introductions should prioritize accessibility and compelling narrative over technical complexity. Authors should "write for the generalist, not for the specialized reader" as this approach "will compel you to take a bird's-eye view of your work: what did you achieve exactly? What were the bases for launching this study?" (Stahelin, 1994). The introduction needs to be "much better than catchy. It has to be compelling" by "being distinctive" and identifying "what is unique, not so much with your research but with yourself, your personality, your voice" (Stahelin, 1994).

The opening deserves special attention, as "the opening paragraph should be your best paragraph, and its opening sentence should be your best sentence" (Chopra et al., 2021) (Goldreich, 2000). Effective openings can employ various strategies, including "an interesting story, vivid description of a scene, using a strong quotation or giving some intriguing facts or making some controversial statements" (Jawaid et al., 2019).

At the sentence level, authors should structure their writing to match reader expectations. This means placing "in the topic position (that is, at the beginning of the sentence) the old information linking backward to the previous discussion" and placing "in the stress position (that is, at the end of the sentence) the new information you want to emphasize" (Chopra et al., 2021). Additionally, authors should "place the subject of the sentence in the topic position, and follow the subject with the verb as soon as possible" and "express the action of each sentence in its verb" (Chopra et al., 2021).

Paragraph structure should serve as the fundamental unit of organization. Each paragraph should "begin with a sentence that summarizes the topic to be discussed or with a sentence that helps the transition from the previous paragraph" and "provide a context for the discussion before asking the reader to consider new information" (Chopra et al., 2021).

Brevity remains essential throughout the writing process. Authors should "condense your introduction into no more than two or three paragraphs, one page maximum in toto" while ensuring "that the writing is fluid and clear" because "brevity is golden" (Stahelin, 1994). The ideal length is typically "2 to 4" paragraphs that "briefly contextualize the science and the problem faced, showing how important this specific study is to the field" (Neto et al., 2023). However, introductions for interdisciplinary work may justify additional length when authors need to "write about different areas and concepts" (Neto et al., 2023).

Evidence

(Stahelin, 1994)

Guidelines

"Write for the generalist, not for the specialized reader. This will compel you to take a bird's-eye view of your work: what did you achieve exactly? What were the bases for launching this study? Which will stand as your most robust conclusion? What are others in the field likely to pursue in direct filiation?"

"your introduction needs to be much better than catchy. It has to be compelling. How can you achieve it? By being distinctive. Identify what is unique, not so much with your research but with yourself, your personality, your voice"

"(i) Spell out and make explicit what is a given to you. It is one of the functions of an introduction to tell why you did this particular study and what you hoped to achieve. (ii) Anchor your contribution into the scientific flow, i.e., refer to earlier work by others and to contemporary work by your competitors"

"Don't omit telling the readers what they are about to find in the rest of your paper, if they keep at it. Your introduction provides an advance summary"

"Condense your introduction into no more than two or three paragraphs, one page maximum in toto. In making it short and crisp, make sure that the writing is fluid and clear. Brevity is golden."

(Chopra et al., 2021)

DeepABM: Scalable, efficient and differentiable agent-based simulations via graph neural networks

"In writing the introduction, you should remember the following maxim:

The opening paragraph should be your best paragraph, and its opening sentence should be your best sentence. (Knuth, Larrabee,

and Roberts 1989, 5) You cannot achieve such an ambitious goal on the first try; instead as you add new sections to the paper, you should review and revise all sections written so far. For more on the spiral plan of writing, see pages 131-133 of (Erickson, 2010).

a. Like the abstract, the introduction should be accessible to general readers in the field of simulation.

b. For methodology papers and advanced tutorials, substantially more advanced background may be assumed in the sections following the introduction"

"In constructing each sentence, place old and new information in the respective positions where readers generally expect to find such information. For an excellent discussion of the principles of scientific writing based on reader expectations, see (Schlesinger, 1979) and Williams and Bizup (2014, 2017). a. Place in the topic position (that is, at the beginning of the sentence) the old information linking backward to the previous discussion.

b. Place in the stress position (that is, at the end of the sentence) the new information you want to emphasize.

c. Place the subject of the sentence in the topic position, and follow the subject with the verb as soon as possible.

d. Express the action of each sentence in its verb"

"Make the paragraph the unit of composition. a. Begin each paragraph with a sentence that summarizes the topic to be discussed or with a sentence that helps the transition from the previous paragraph.

b. Provide a context for the discussion before asking the reader to consider new information."

(Goldreich, 2000)

How to write a paper

"I. Pre-Writing/Information Gathering Step At the start of the writing process, the writer needs to make sure that he/she understands what his/her teacher's expectations are for this particular assignment. Then he/she needs to develop a topic, focus statement (thesis) and do research A. Brainstorming This is a great step that helps get the brain going. Basically in a list format write down all the ideas that come into the writer's mind. Brainstorming can be useful in generating ideas for a paper as well as for points to support and explain each point. Below is an example of a student's brainstorming list over possible topics for a health sciences course research paper:"

(Jawaid et al., 2019)

How to write introduction and discussion

"It is essential that the introduction is brief, it should be well referenced. It should provide an outline of the current thinking, identify the gaps which exist in knowledge which the author wishes to fill. (1973) Some basic requirements for writing introduction include that it is not only short but also arresting. It should tell the readers what to say, is it worth saying. Is it the right format for conveying this message? Tell the readers why you have undertaken to conduct this study. What is already known on this subject and what new information will be added to the medical literature. (Goldreich, 2000)"

"The best introduction should include information from systemic review of work already done before and also demonstrate that what more work needs to be done. The objective is to answer a question and one should undertake the study only if the question cannot be answered from the available literature. After doing proper literature search, select only a few latest studies pertaining to mostly last five years. Make sure that you have read important studies already undertaken on that particular subject and they should be referenced in this section. If there are about twenty five references in the whole manuscript, make sure that four to five or at the most one third of the references are covered in the introduction. The possible start for an introduction can be an interesting story, vivid description of a scene, using a strong quotation or giving some intriguing facts or making some controversial statements. [4] Introduction must also describe the magnitude of the problem which should be followed by brief description of current knowledge and the gap which still exists. [5] At the end of the introduction, restate the objective of the study."

(Neto et al., 2023)

The Art of Scientific Writing and Ethical Use of Artificial Intelligence

"The most important and most neglected characteristic of the introduction is that it must not be a review. Some manuscripts have such long introductions with so many distractions that readers -those whom you worked so hard to attract in the title and abstract - quit reading the manuscript while struggling through the introduction"

"beginners often believe they should start introductions with "J. J. Thomson's discovery of the electron..." and go on to craft a chronological narrative until they have fully explained the current knowledge in the field. But the best introductions, rather than providing such historical reviews, should briefly contextualize -in a few paragraphs, let us say 2 to 4 -the science and the problem faced, showing how important this specific study is to the field"

"well-written introductions should do more: they should have a "magic" effect on readers, making them feel captivated by the science and the advancements that the study promises to offer, thus motivating overall engagement with the manuscript"

"Regarding length, introductions should never be more than a few paragraphs (2 or 3 at least and 6 or 7 at most), regardless of the type of manuscript. But the length also depends on many factors and may vary accordingly. If you are writing a full paper in an interdisciplinary field -for instance, when you have monitored a multicomponent reaction with a new mass spectrometric technique - you likely need to write about different areas and concepts; therefore, a longer introduction is justified."

Common Mistakes and What to Avoid

The most common mistakes in scientific introductions include writing extensive literature reviews instead of focused context, creating chronological historical narratives, making introductions too long, and failing to clearly state research objectives and significance. (5 sources)

Avoid turning your introduction into a literature review. One of the most serious errors is attempting "to do an extensive literature review in the introduction, which often goes off-topic" ([Zelst et al., 2022](#)). Remember that "the most important and most neglected characteristic of the introduction is that it must not be a review" ([Neto et al., 2023](#)). Such extensive reviews create "such long introductions with so many distractions that readers -those whom you worked so hard to attract in the title and abstract -quit reading the manuscript while struggling through the introduction" ([Neto et al., 2023](#)).

Don't start with chronological historical narratives. Beginners commonly "believe they should start introductions with 'J. J. Thomson's discovery of the electron...' and go on to craft a chronological narrative until they have fully explained the current knowledge in the field" ([Neto et al., 2023](#)). Instead of providing "a full history of the topic," only quote "previous work having direct bearing on the present problem" ([Haswani et al., 2023](#)).

Avoid making your introduction too long. Don't "over-burden the reader by making the introduction too long" and instead "get to the key parts of the paper sooner rather than later" ([Haswani et al., 2023](#)). The introduction should be "brief" and "well referenced" while providing "an outline of the current thinking" ([Jawaid et al., 2019](#)).

Don't write for specialists only. A common mistake is failing to "write for the generalist, not for the specialized reader" ([Stahelin, 1994](#)). Authors should also avoid failing to "spell out and make explicit what is a given to you" since readers may not share your specialized knowledge ([Stahelin, 1994](#)).

Avoid creating subsections within the introduction. Authors should "avoid to make a sub section in Introduction" ([Haswani et al., 2023](#)) as this disrupts the flow and coherence of the section.

Don't forget to state your research objectives clearly. Many authors fail to adequately address fundamental questions like "what is the problem to be solved? What previous work has been done? What is its main limitation? What do you hope to achieve?" ([Zelst et al., 2022](#)). The "research objective is the most important part of the introduction" ([Haswani et al., 2023](#)).

Paper	Target audience	Scope of guidance	Inclusion of best practices	Emphasis on clarity and conciseness	Domain specificity	Practical examples provided
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Paper	Target audience	Scope of guidance	Inclusion of best practices	Emphasis on clarity and conciseness	Domain specificity	Practical examples provided
101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth	First-time and experienced modellers, collaborators, reviewers	Entire geodynamic modelling process, from equations to communication	Highlights best practices and discusses their implementations	Clarifies and consolidates terminology for clear communication	Tailored to geodynamic modelling and solid Earth sciences	Ample examples from lithosphere and mantle dynamics.
The Art of Scientific Writing and Ethical Use of Artificial Intelligence	Less experienced scientific writers, beginners, and students.	The paper covers the entire scientific writing process, from initial drafting to submission and publication, including specific sections and ethical AI use.	Yes, the paper offers crucial tips, concepts, and best practices for effective scientific writing, covering various stages from initial drafting to publication.	The paper heavily stresses clear, comprehensible, and concise language as crucial for effective scientific communication and manuscript impact.	The writing advice provided is general to all scientific papers, not tailored to specific domains.	The paper includes concrete examples of both good and bad scientific writing practices, particularly for sentences, titles, and abstracts.
Integrating extensive and intensive reading worksheets on digital platforms	Teachers, education practitioners, reading enthusiasts, and extensive/intensive reading associations.	The paper provides guidance on writing the introduction section of a research manuscript.	Yes, the paper provides actionable advice for writing effective scientific paper introductions.	The paper strongly emphasizes conciseness and clarity in scientific writing, especially for introductions and instructional materials.	The writing advice is general to all scientific papers, not tailored to specific scientific domains.	The paper provides general strategies and types of activities for worksheets but lacks concrete examples of specific questions or tasks.

Paper	Target audience	Scope of guidance	Inclusion of best practices	Emphasis on clarity and conciseness	Domain specificity	Practical examples provided
Guidelines	beginners, experienced researchers, reviewers	Abstract writing	Yes, paper provides actionable advice and recommended methods.	Strongly stresses clarity, conciseness, and simple language.	General to all scientific papers	Yes, includes both good and bad examples.

Evidence

(Zelst et al., 2022)

[101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth](#)

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(Haswani et al., 2023)

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(Stahelin, 1994)

Guidelines

"Write for the generalist, not for the specialized reader. This will compel you to take a bird's-eye view of your work: what did you achieve exactly? What were the bases for launching this study? Which will stand as your most robust conclusion? What are others in the field likely to pursue in direct filiation?"

"your introduction needs to be much better than catchy. It has to be compelling. How can you achieve it? By being distinctive. Identify what is unique, not so much with your research but with yourself, your personality, your voice"

"(i) Spell out and make explicit what is a given to you. It is one of the functions of an introduction to tell why you did this particular study and what you hoped to achieve. (ii) Anchor your contribution into the scientific flow, i.e., refer to earlier work by others and to contemporary work by your competitors"

"Don't omit telling the readers what they are about to find in the rest of your paper, if they keep at it. Your introduction provides an advance summary"

"Condense your introduction into no more than two or three paragraphs, one page maximum in toto. In making it short and crisp, make sure that the writing is fluid and clear. Brevity is golden."

Length and Organization Guidelines

Scientific introductions should be 2-4 paragraphs (roughly one page) and about ten times longer than the abstract, with longer introductions justified only for interdisciplinary work. The content should be self-contained and move systematically from general context to specific research objectives. (5 sources)

The optimal length for scientific introductions follows well-established guidelines that balance comprehensiveness with reader engagement. Introductions should typically be "about ten times longer as a rough approximation" compared to the abstract while remaining "sufficiently self-contained, so that important readers and evaluators do not need to read the rest of text, being assured in the contribution made, and validity of the text to follow" ([Stojmenovic et al., 2012](#)). In practical terms, this translates to condensing "your introduction into no more than two or three paragraphs, one page maximum in toto" ([Stahelin, 1994](#)).

The paragraph count should generally range from "2 or 3 at least and 6 or 7 at most" with the understanding that "2 to 4" paragraphs should "briefly contextualize the science and the problem faced, showing how important this specific study is to the field" ([Neto et al., 2023](#)). However, length flexibility is appropriate for certain contexts: "If you are writing a full paper in an interdisciplinary field -for instance, when you have monitored a multicomponent reaction with a new mass spectrometric technique -you likely need to write about different areas and concepts; therefore, a longer introduction is justified" ([Neto et al., 2023](#)).

Organizational principles should prioritize efficiency and reader accessibility. Authors must "not over-burden the reader by making the introduction too long" and should "get to the key parts of the paper sooner rather than later" ([Haswani et al., 2023](#)). The organization should "move from general to specific: from the problem in the real world to the literature to your research" while avoiding subsections within the introduction ([Haswani et al., 2023](#)).

For reference management within introductions, authors should be selective in their citations. When working with manuscripts containing "about twenty five references in the whole manuscript, make sure that four to five or at the most one third of the references are covered in the introduction" and focus on "only a few latest studies pertaining to mostly last five years" ([Jawaid et al., 2019](#)). The introduction should remain "brief" and "well referenced" while providing "an outline of the current thinking" without becoming overly comprehensive ([Jawaid et al., 2019](#)).

Evidence

(Stojmenovic et al., 2012)

[How to write research articles in computing and engineering disciplines](#)

"In brief, the introduction of a paper intended to be published in a journal or conference, or Chapter 1 of a master or doctoral thesis, should present the same content (summary of the article), in the same order, as the abstract, with more space provided. Normally it is about ten times longer as a rough approximation"

"In our view, the introduction or Chapter 1 should present sufficient information, and be sufficiently self-contained, so that important readers and evaluators do not need to read the rest of text, being assured in the contribution made, and validity of the text to follow."

(Stahelin, 1994)

[Guidelines](#)

"Write for the generalist, not for the specialized reader. This will compel you to take a bird's-eye view of your work: what did you achieve exactly? What were the bases for launching this study? Which will stand as your most robust conclusion? What are others in the field likely to pursue in direct filiation?"

"your introduction needs to be much better than catchy. It has to be compelling. How can you achieve it? By being distinctive. Identify what is unique, not so much with your research but with yourself, your personality, your voice"

"(i) Spell out and make explicit what is a given to you. It is one of the functions of an introduction to tell why you did this particular study and what you hoped to achieve. (ii) Anchor your contribution into the scientific flow, i.e., refer to earlier work by others and to contemporary work by your competitors"

"Don't omit telling the readers what they are about to find in the rest of your paper, if they keep at it. Your introduction provides an advance summary"

"Condense your introduction into no more than two or three paragraphs, one page maximum in toto. In making it short and crisp, make sure that the writing is fluid and clear. Brevity is golden."

(Neto et al., 2023)

[The Art of Scientific Writing and Ethical Use of Artificial Intelligence](#)

"The most important and most neglected characteristic of the introduction is that it must not be a review. Some manuscripts have such long introductions with so many distractions that readers -those whom you worked so hard to attract in the title and abstract - quit reading the manuscript while struggling through the introduction"

"beginners often believe they should start introductions with "J. J. Thomson's discovery of the electron..." and go on to craft a chronological narrative until they have fully explained the current knowledge in the field. But the best introductions, rather than providing such historical reviews, should briefly contextualize -in a few paragraphs, let us say 2 to 4 -the science and the problem faced, showing how important this specific study is to the field"

"well-written introductions should do more: they should have a "magic" effect on readers, making them feel captivated by the science and the advancements that the study promises to offer, thus motivating overall engagement with the manuscript"

"Regarding length, introductions should never be more than a few paragraphs (2 or 3 at least and 6 or 7 at most), regardless of the type of manuscript. But the length also depends on many factors and may vary accordingly. If you are writing a full paper in an interdisciplinary field -for instance, when you have monitored a multicomponent reaction with a new mass spectrometric technique - you likely need to write about different areas and concepts; therefore, a longer introduction is justified."

(Haswani et al., 2023)

[Integrating extensive and intensive reading worksheets on digital platforms](#)

"Keep your Introduction to be very concise, well structured, and inclusive of all the information needed to follow the development of

your findings. Do not over-burden the reader by making the introduction too long. Get to the key parts of the paper sooner rather than later. Best Practice: (1) begin the introduction by providing a concise background account of the problem studied; (2) state the objective of the investigation. Your research objective is the most important part of the introduction; (3) establish the significance of your work: why was there a need to conduct the study?; (4) introduce the reader to the pertinent literature. Do not give a full history of the topic. Only quote previous work having direct bearing on the present problem; (5) clearly state your hypothesis, the variables investigated, and concisely summarize the methods used; (6) define any abbreviations or specialized/regional terms; (7) provide a concise discussion of the results and findings of other studies so the reader understands the big picture; (8) describe some of the major findings presented in your manuscript and explain how they contribute to the larger field of research; (9) state the principal conclusions derived from your results, and; (10) identify any questions left unanswered and any new questions generated by your study. Be concise and aware of who will be reading your manuscript and make sure the Introduction is directed to that audience. Move from general to specific: from the problem in the real world to the literature to your research. Last, please avoid to make a sub section in Introduction."

(Jawaid et al., 2019)

How to write introduction and discussion

"It is essential that the introduction is brief, it should be well referenced. It should provide an outline of the current thinking, identify the gaps which exist in knowledge which the author wishes to fill. (1973) Some basic requirements for writing introduction include that it is not only short but also arresting. It should tell the readers what to say, is it worth saying. Is it the right format for conveying this message? Tell the readers why you have undertaken to conduct this study. What is already known on this subject and what new information will be added to the medical literature. (Goldreich, 2000)"

"The best introduction should include information from systemic review of work already done before and also demonstrate that what more work needs to be done. The objective is to answer a question and one should undertake the study only if the question cannot be answered from the available literature. After doing proper literature search, select only a few latest studies pertaining to mostly last five years. Make sure that you have read important studies already undertaken on that particular subject and they should be referenced in this section. If there are about twenty five references in the whole manuscript, make sure that four to five or at the most one third of the references are covered in the introduction. The possible start for an introduction can be an interesting story, vivid description of a scene, using a strong quotation or giving some intriguing facts or making some controversial statements. [4] Introduction must also describe the magnitude of the problem which should be followed by brief description of current knowledge and the gap which still exists. [5] At the end of the introduction, restate the objective of the study."

Tools and Resources

Modern scientific writing benefits from structured online guidelines and AI tools that can help researchers organize ideas and create compelling introductions. These resources provide specific criteria for evaluation and assistance with structuring content effectively. (2 sources)

Several practical tools and resources are available to support researchers in crafting effective scientific introductions. Online writing platforms like Scribbr provide comprehensive guidelines specifically tailored for scientific paper writing, offering structured criteria for evaluating introduction quality (Yuan et al., 2024). These resources typically include specific evaluation aspects such as "Introduce Your Topic: Does the introduction effectively identify the subject matter and provide sufficient background to inform the reader about the topic being addressed?" which help authors self-assess their work (Yuan et al., 2024).

Artificial intelligence tools have emerged as valuable resources for scientific writing support. AI platforms like ChatGPT can assist researchers in "structuring and arranging the ideas the researcher wishes to communicate in the introductory paragraph, ultimately assisting in the creation of a clear and engaging thesis statement" (Uranbey et al., 2024). These tools are particularly useful for helping authors define research objectives and emphasize the significance of their studies, which can "enhance the reader's comprehension of the study's purpose and its broader importance" (Uranbey et al., 2024).

Beyond digital tools, traditional resources remain valuable for scientific writing guidance. Style guides from major scientific publishers, writing handbooks specific to scientific disciplines, and institutional writing centers provide established frameworks and examples (Model-Generated). Many universities also offer workshops and courses focused specifically on scientific writing skills, while peer review groups and writing collaboratives can provide feedback on draft introductions before submission (Model-Generated).

Evidence

(Yuan et al., 2024)

[LLMCrit: Teaching Large Language Models to Use Criteria](#)

"We sourced our guidelines for scientific paper writing from the Scribbr website.¹ From this resource, we manually extracted specific aspects tailored for the introduction part of a scientific paper. As a result, we curated five aspects. We used the prompt in Appendix Tab. 5 for criteria extraction, which resulted in 11 criteria in total. An example criterion of this task is "Introduce Your Topic: Does the introduction effectively identify the subject matter and provide sufficient background to inform the reader about the topic being addressed?"."

(Uranbey et al., 2024)

[Assessing ChatGPT's Diagnostic Accuracy and Therapeutic Strategies in Oral Pathologies: A Cross-Sectional Study](#)

"ChatGPT can represent a valuable resource for crafting a scientific paper's introductory paragraph. This initial section serves the crucial purpose of defining the research objectives and underlining the study's significance. ChatGPT can be of assistance in structuring and arranging the ideas the researcher wishes to communicate in the introductory paragraph, ultimately assisting in the creation of a clear and engaging thesis statement. This, in turn, will enhance the reader's comprehension of the study's purpose and its broader importance."